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CASE-STUDY-1

BrightMart is a medium sized online shopping company customes" used its website and mobile application to purchase products.

The company stores

i) customes names
ii) Email Address
iii) Phone Numbers
iv) Delivery Addresses
v) Passwords
vi) order Information
vii) Payment-related information

BrightMart has around 100 employees.

The company recently hired several new employes. To make things easier, the IT department gave employees acces to different internal system

However, the comland has the following security weakness:

- Some employee use simple passwords such as BrightMart123.
- Many employees use the same Password for multiple ~~user~~ systems.

The Incident

one Monday morning, an employee named Rahul receives an email.

Subject

- ~~Subject~~
- Software updates are sometimes delayed.
- The company does not regularly conduct employee cybersecurity training.
- There is no multi-factor authentication for some important accounts.
- Employees can access sensitive information tha they do not necessarily need for their jo
- Backups are performed only once a month

The Incident

one Monday morning, an employee named Rahul receives an email.

Subject: Urgent - Your Account will Be Suspended

The email appears to come from the company's IT department.

It says:

"Your company account will be suspended today. Click the link below and verify your password immediately."

Rahul clicks the link.

The page looks almost identical to the company's login page.
He enters his username and password.
Rahul does not realize that the website is controlled by an attacker.
The attacker now has Rahul's login credentials.

Stage 1 - Unauthorized Access

The attacker uses Rahul's credentials to access BrightMart's internal system.
Because Rahul uses the same password for another system, the attacker is also able to access that system.
The attacker discovers that Rahul has access to customer information.
The attacker downloads a large amount of customer data.

Question 3: What cybersecurity problem has occured

which ... attacker to move further into the organization?

Stage 2 - Malware Attack

The attacker sends another email to Rahul containing a malicious attachment.
Rahul opens the attachment
Malware is installed on his computer.
The Malware begins communicating with the attacker's system.
The attacker now attempts to move from Rahul's computer to other system within BrightMart's network.

2) Question :-
Which vulnerability allowed the attacker to move further into the organization?

Stage 3 - Ransomware

Two days later, employees arrive at work and discover that they cannot access important files.
A message appears on their computers:

"Your files have been encrypted. Pay the ransom to recover them."

The company's customer database and several important business files are encrypted.
Unfortunately, the Company's most recent back

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is almost a month old.
BrightMart now faces several Problems.
BrightMart website stops working.
Financial Customers cannot Place orders.
The customer-support team cannot access customer records.

STAGE 4- Business Impact

BrightMart now faces several Problems.

Financial Impact:-
The company loses sales because its website is unavailable.

Data Impact :-
Customer information may have been stolen.

Operational Impact:-
Employees cannot access important systems.

Reputation Impact:-
customers lose confidence in the company.

Legal/compliance Impact :-
The company may face regulatory or legal consequences depending on what information was exposed and

which laws apply.

Group 1 — Identify the Threats

3) Question

What threats are present in this case?

Group 2 - Identify the vulnerabilities

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Answer 2) Possible answers:
- Phising
- Credential theft
- Unauthorized access
- Data breach

Ans 2) Student should identify possibilities such as
- Weak Passwords
- Excessive users privileges
- Lack of MFA
- Poor network security
- Lack of employee awareness
- outdated software

Ques) Ans 3) what threats are present in this case?

→ They may identify:
- Phising
- Malware
- Ransomware
- Credential theft
- Unauthorized access
- Data theft

Group-2
Q.4) what weaknesses in BrightMart's security allowed the attack to succeed?

→ Expected answer:

Vulnerability	How it helped the attacker
Weak passwords	Easier to compromise accounts
Password reuse	one stolen password gives access to multiple systems.
No MFA	stolen password may be enough to log in
Lack of training	Employee falls for phising
Delayed updates	vulnerable software may be exploited
Excessive Privileges	Attacker can access more data
Infrequent backups	Difficult to cover after ransomware

Ask students to determine how the attacker affected:

3) Apply the CIA Triad

1) Confidentiality:

Q Was confidential information exposed?

→ Yes.

customer information was accessed / downloaded by the attacker

2) Integrity:

Q Was information changed or damaged?

→ Potentially Yes.
The attacker could modify or encrypt files.

3) Availability:

Q Where system available to legitimate users?

→ No.
The website and internal systems became unavailable because of ransomware.

So:
Confidentiality → Affected
Integrity → Affected
Availability → Affected

Group 4 — Security Solutions

Q.1) Who is responsible for cybersecurity?

What should BrightMart do to prevent this from happening again

→ Expected discussion:

Not only IT department. Every employee who uses the organization's systems has a role in cybersecurity.

Q.2) was phising the only problem?

→ No.
Phising was the initial attack but weakness su as Password reuse, lack of MFA, excessive priv , Poor backups and insufficient training allow the attack to become much more damaging.

Q.3) what was the biggest vulnerability?

→ There is no single correct answer.
Students should justify their choice.
For example:
"Lack of ~~backup~~ employee awareness was was major vulnerability because the attack started when Rahul clicked the phising link".
Another student might argue:
"Poor backup practices were the biggest pr lem because BrightMart could not quickly recover from ransomware".
Both can be valid if properly justified.

What should BrightMart do to prevent this from happening again?

→ ~~Should~~ students could suggest:

- Use strong passwords.
- Implement Multi-factor Authentication (MFA).

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- Provide regular cybersecurity awareness training.
- Conduct Phising-awareness exercises.
- Keep software updated.
- Follow the principle of least privilage.
- segment important networks.
- Maintain regular backups.
- Encrypt sensitive data.
- Deploy antivirus/endpoint protection.
- Monitor suspicious login activity.
- Develop an incident-response plan.

Done
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